

Modes of Operation

Function	FM mode details	AM mode details
Input frequency range	11 to 150 MHz	0.5 to 30 MHz
Prescaler	In operation	Not in operation
Input voltage	300mV pk-pk recommended	300mV pk-pk recommended
Frequency resolution	10 kHz: 11.00 to 99.99 MHz 100 kHz: 11.0 MHz to 149.9 MHz Set in radio to 100kHz resolution	1 kHz: 500 to 9999 kHz 10 kHz: 0.50 to 29.99 MHz Set in radio to 10kHz resolution for SW and 1kHz resolution for MW
IF offset	10.7 MHz 70 kHz also available	455 kHz
Input resistance	150 ohms	2000 ohms

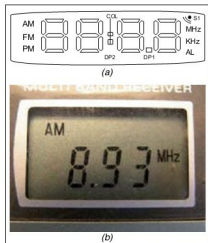


Fig. 2: (a) 4 digit-LCD and (b) image of the LCD

in the FM mode. In the AM mode, it is 455 kHz above the displayed frequency. 70kHz offsets are also available as an option in the FM mode, but we are skipping this for now as the radios we rip these modules from have no 70kHz IF.

The LCD driver converts binary data into voltages necessary to drive the LCD unit.

Opening up the radio

The best part about Kchibo KK-939B radio is use of minimal external points. Also, it is simple to understand and operate, has an LED light to illuminate the LCD, and looks good. I recommend beginners to buy this radio and rip it apart for a clean, high-performance frequency counter.

To access the frequency counter board, the radio (Kchibo KK-939B) has to be opened up. In the close-up of the



Fig. 3: The radio



Fig. 4: Opened up radio; note that the brown PCB is the frequency counter and LCD section and green PCB is the radio receiver

frequency counter PCB (Fig. 9), you will notice the solder joints marked VH input, HF input, points '1' through '5,' FM and AM. VHF input is to be connected to the RF source whose frequency you want to determine (from 11 to 150 MHz). HF input is to be connected to the RF source whose frequency you



Fig. 5: The brown PCB with the metallic shield, which is our target



Fig. 6: Opened up: The frequency counter UTC6610 hidden away behind the LCD, while the flip-flop circuitry (just to set FM, AM states) is visible as the SMD transistors



Fig. 7: The back side: RF shield-cum-LCD mechanical support. Notice the purple and black wire for VHF and HF input, respectively



Fig. 8: The purple line leads to the take-off point for the VHF oscillator (FM section), while the rightmost black wire leads to the take-off point for the AM oscillator

want to determine (up to 30 MHz).

'1' is an input point. It has to be kept at 0V for FM (VHF) operation and 10kHz resolution in the AM mode